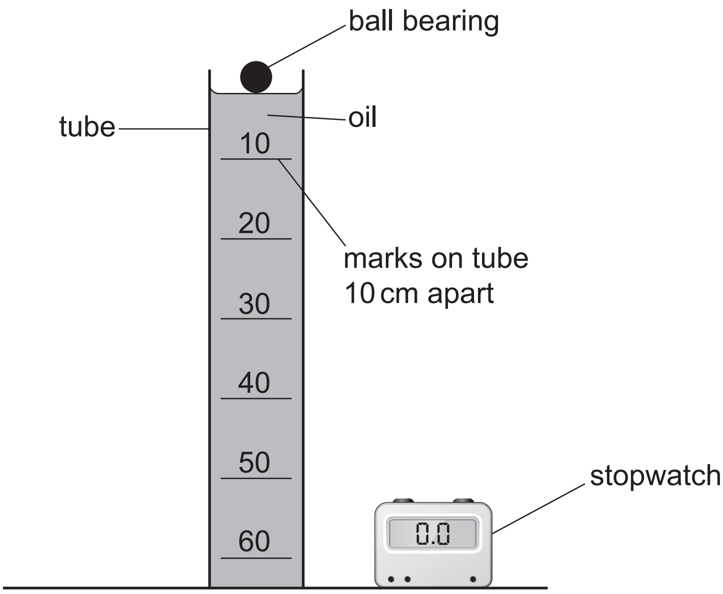
2. Students investigate the terminal speed of a ball bearing in oil.
They measure the time it takes for the ball bearing to drop different distances through the oil.



The results from the experiment are shown in the table below.

Distance (cm)	Time (s)			
	Trial 1	Trial 2	Trial 3	Mean
10	5.4	6.2	5.8	5.8
20	7.6	4.2	8.0	7.8
30	8.4	9.0	8.3	8.6
40	10.9	10.3	10.4	10.5
50	11.5	11.2	10.8	11.2
60	12.5	13.0	13.2	12.9

(a) (i) Freya states that the mean time for 20 cm is incorrect and should be 6.6 s.
Explain whether Freya is correct. [2]

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Examiner only

(ii) Evaluate the repeatability of the data. [1]

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(b) State **one** source of inaccuracy in this method **and** how it could be reduced. [2]

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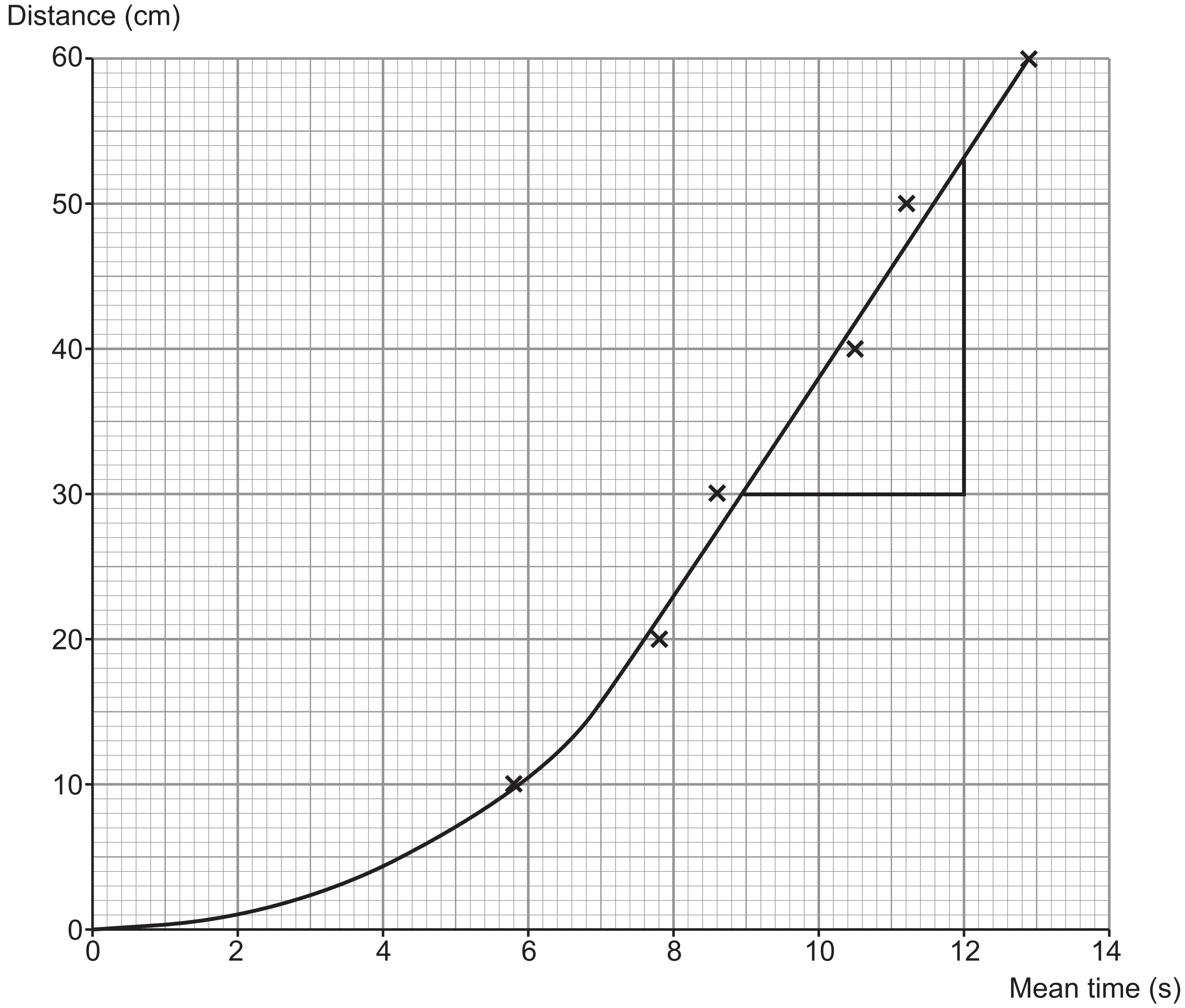
(c) A graph of the students' data is given below.



Examiner only

The section of the graph where the line is straight represents the ball bearing travelling at terminal speed.

(i) Estimate the time at which the ball bearing reaches terminal speed. [1]

time = ... s

(ii) The gradient of a distance-time graph represents the speed of an object. Use the equation:

speed = gradient of distance-time graph

and the triangle shown on the graph to calculate the terminal speed of the ball bearing. [2]

terminal speed = ... cm/s

(iii) State how the **acceleration** of the ball bearing changes as it falls through the oil. [2]

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